1. K-163 recovery transfer and horizontal control

Formal gates, not a weakness result · 2026-10-31

Outcome. Let r = 5846006549323611672814741753598448348329118574063. The compiled Lean module checks gcd(45641, r) = 1. A generic theorem proves that additive maps with recovery(phi(X)) = [45641]X make phi injective on r-torsion, preserve exact point order, and preserve and reflect scalar relations.

[r]Q=0 says only that Q is r-torsion. Calling the scalar-relation equivalence a discrete-log result requires a separate proof that Q belongs to the cyclic subgroup generated by the base P.

A stronger same-curve control defines

$$\text{alpha} = 51 - 160\,\tau,\ \text{alpha_bar} = -109 + 160\,\tau$$

and derives both compositions [45641] from tau^2 = tau - 2. This proves no vertical map, vertical dual, concrete Frobenius bridge, speedup, attack, or weak curve. Both named targets compile and pass requested-theorem axiom audits. The repository-wide receipt remains pending because an unrelated pre-existing Semaev certificate exceeded local memory.

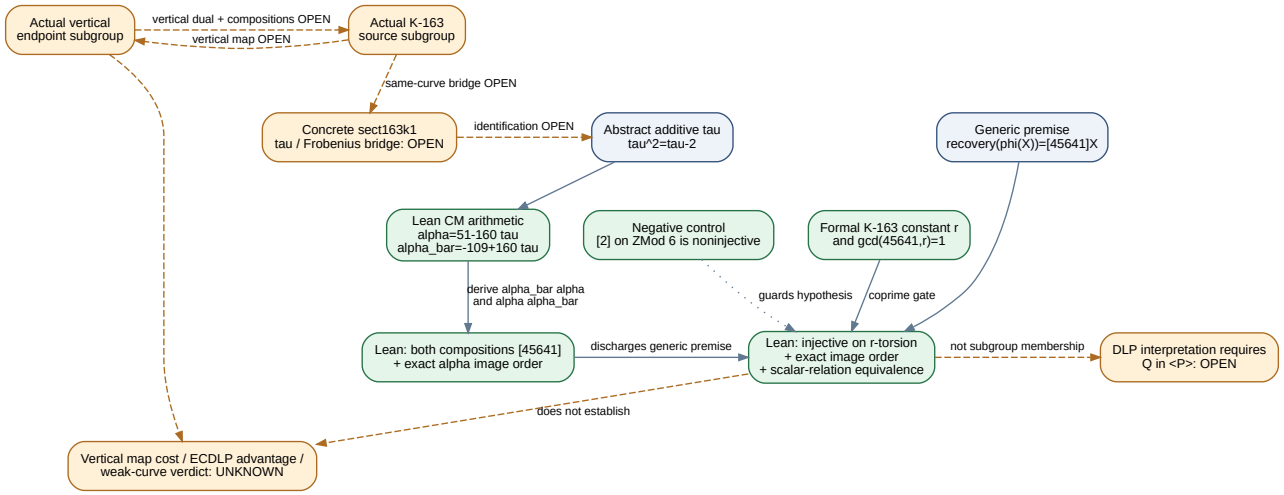


Figure 1: Green is closed or conditional formal scope. Amber dashed edges are open geometric, membership, and computational bridges.

1.1. Proof and controls

For a generic collision put Z=P-Q. Recovery gives [45641]Z=0; the two torsion hypotheses give [r]Z=0. Bezout coefficients for the coprime pair then give Z=0. Multiplication by 2 on ZMod 6 is the negative control: it has the recovery form but maps both 0 and 3 to 0.

Conjugation sends tau to 1-tau, so conj(51-160 tau)=-109+160 tau. Using tau^2=tau-2, the constant coefficient of the product is -5559+51200=45641 and the tau coefficient is 8160+17440-25600=0. Lean also derives both corresponding point-action compositions and exact image order.

Closed integer arithmetic also matches the two archived subgroup residues and their product 45641 mod r; this is conditional on the external statement that concrete Frobenius acts through the published eigenvalue.

1.2. Transfer ledger

Obligation	State	Meaning
K-163 constant and coprimality	closed	kernel arithmetic in the compiled module
Generic recovery implication	conditional	recovery o phi = [45641] is an input
Horizontal compositions	derived	both orders follow from the abstract tau relation
Archived scalar residues	closed integers	values and product match; concrete Frobenius eigenvalue bridge remains open
Exact image order	derived	an order-r point remains order r
DLP membership	open	[r]Q=0 does not imply Q in <P>
Concrete tau / Frobenius	open	the actual sect163k1 map is not identified in Lean
Vertical map and dual	open	no retained degree-45,641 vertical point maps
Vertical compositions	open	horizontal identities do not certify them
Cost and ECDLP advantage	unknown	no executable vertical transfer to price
Weak-curve assertion	unsupported	no lower work factor or key recovery

1.3. Exact boundary

The horizontal result is a same-group control, not a conductor descent. Its abstract tau still needs a concrete Frobenius bridge. The weakest vertical obligation remains an explicit independently checkable point map plus recovery/dual and a source composition certificate. Only then should the project run planted scalar round trips and fully charge construction, preprocessing, evaluation, verification, and amortization.

The producer-authored hypothesis table is not independent semantic review. A “breakthrough” classification would require independent review of the mathematics, concrete bridges, and complete cost claim.